

OASIS INFOBYTE
DATA ANALYTICS INTERNSHIP
LEVEL 2 – TASK 2
WINE QUALITY PREDICTION

Final Project Report

Submitted By

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Internship

OASIS INFOBYTE – Data Analytics Internship

Domain

Data Analytics / Machine Learning

Project Title

Level 2 – Task 2: Wine Quality Prediction

Technologies Used

Python • Pandas • NumPy • Matplotlib • Seaborn • Scikit-learn • Jupyter Notebook

Submitted To

OASIS INFOBYTE

Year

2026

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to OASIS INFOBYTE for providing me with the opportunity to complete Level 2 – Task 2, Wine Quality Prediction. This project provided practical exposure to exploratory data analysis, data quality assessment, feature analysis, binary classification, train-test splitting, feature scaling, model training, classification evaluation, confusion-matrix analysis, model comparison, and feature-importance interpretation.

I also thank the mentors and coordinators for providing the internship task requirements and learning resources. The structured workflow helped me understand how physicochemical wine measurements can be transformed into a machine-learning classification problem and how different classification algorithms can be compared using consistent evaluation criteria.

The implementation was completed in a Jupyter Notebook using the red wine quality dataset. The workflow begins with dataset loading and inspection, continues through exploratory analysis and binary target construction, and ends with Random Forest, SGD Classifier, and Support Vector Classifier evaluation.

This work strengthened my practical understanding of supervised machine learning and demonstrated the importance of data quality, class definition, feature scaling, appropriate evaluation metrics, and interpretable model comparison in classification problems.

ABSTRACT

This project focuses on predicting wine quality using classification algorithms based on physicochemical properties of red wine. The dataset contains 1,599 wine samples and 12 columns, including 11 numerical chemical features and the original quality score. The workflow begins with dataset loading, structural inspection, descriptive statistics, missing-value analysis, quality-score distribution analysis, feature distribution visualization, and correlation analysis.

To convert the quality-prediction problem into a classification task, quality scores below 6 were labelled as Not Good and scores of 6 or above were labelled as Good. This produced 744 Not Good samples and 855 Good samples. The data was divided into 80% training and 20% testing subsets using stratified sampling, resulting in 1,279 training samples and 320 testing samples. StandardScaler was applied for the SGD and SVC models.

Three classifiers were trained and evaluated: Random Forest, Stochastic Gradient Descent (SGD) Classifier, and Support Vector Classifier (SVC). Random Forest achieved the highest test accuracy of 81.56%, followed by SVC at 76.25% and SGD Classifier at 65.63%. Precision, recall, F1-score, and confusion matrices were also examined. Random Forest feature importance identified alcohol, sulphates, and volatile acidity as the three most influential features in the classification model.

The results demonstrate that the implemented workflow successfully transforms wine-quality measurements into a binary classification problem and that Random Forest provided the strongest performance among the three evaluated models.

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CHAPTER 1: INTRODUCTION

1.1 Project Overview

The Wine Quality Prediction project focuses on building supervised classification models that predict whether a red wine belongs to a Good or Not Good quality class based on physicochemical measurements. The workflow starts with loading and inspecting the wine-quality dataset, followed by descriptive analysis, missing-value verification, quality distribution analysis, feature distributions, and correlation analysis.

The implementation uses Python, Pandas and NumPy for data handling, Matplotlib and Seaborn for visualization, and Scikit-learn for train-test splitting, feature scaling, classification modelling, and evaluation. The complete workflow is documented in a Jupyter Notebook.

1.2 Problem Statement

Wine quality is influenced by several measurable chemical properties, including acidity, sulfur dioxide, density, pH, sulphates, and alcohol. The original quality variable is an ordinal score, which can be difficult to predict directly with a small set of conventional classifiers. The project therefore converts the score into two practical classes and evaluates how accurately machine-learning models can distinguish wines below the quality threshold from wines meeting or exceeding it.

1.3 Project Objectives

1. Load and inspect the red wine quality dataset.
2. Perform descriptive statistics and assess data quality and missing values.
3. Analyze the distribution of wine quality scores and chemical features.
4. Study relationships among physicochemical variables using correlation analysis.
5. Convert the original quality score into a binary classification target.
6. Prepare the feature matrix and apply an 80:20 stratified train-test split.
7. Standardize features for SGD and SVC models.
8. Train Random Forest, SGD Classifier, and SVC models.
9. Evaluate classifiers using accuracy, precision, recall, F1-score, and confusion matrices.
10. Compare model performance and identify the most influential chemical features using Random Forest feature importance.

CHAPTER 2: DATASET & TOOLS

2.1 Dataset Description

The project uses the red Wine Quality dataset. The executed notebook reports an initial shape of 1,599 rows and 12 columns. Eleven columns represent physicochemical measurements, while the quality column contains the original wine quality score. The chemical variables include fixed acidity, volatile acidity, citric acid, residual sugar, chlorides, free sulfur dioxide, total sulfur dioxide, density, pH, sulphates, and alcohol.

2.2 Tools & Technologies Used

TOOL / TECHNOLOGY	PURPOSE
Python	Programming language and machine-learning workflow
Pandas	Dataset loading, inspection, transformation, and tabular analysis
NumPy	Numerical operations and supporting calculations
Matplotlib	Plotting and visual analysis
Seaborn	Statistical visualization and heatmaps
Scikit-learn	Train-test split, scaling, classification models, and evaluation
Jupyter Notebook	Interactive implementation and documentation
CSV	Input dataset storage

2.3 Project Folder Structure

The project is organized into separate folders for the dataset, implementation notebook, report documentation, and screenshots. The notebook uses a relative path to load the wine-quality CSV from the dataset folder, while the screenshots folder stores visual evidence of the implementation outputs.

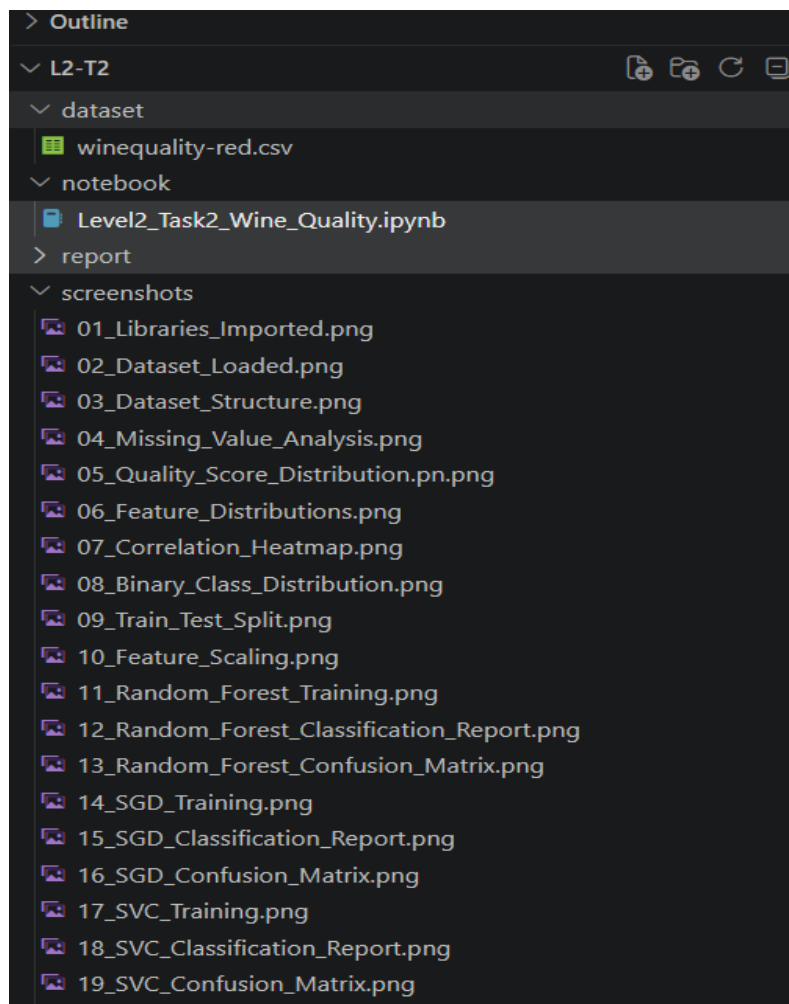


Figure 2.3: Project Folder Structure of the Wine Quality Prediction Project

2.4 Input Dataset

The input file used in the notebook is winequality-red.csv. It is a semicolon-separated CSV containing 1,599 red-wine observations. The quality column is retained as the original score and is later transformed into the binary quality_label used for classification.

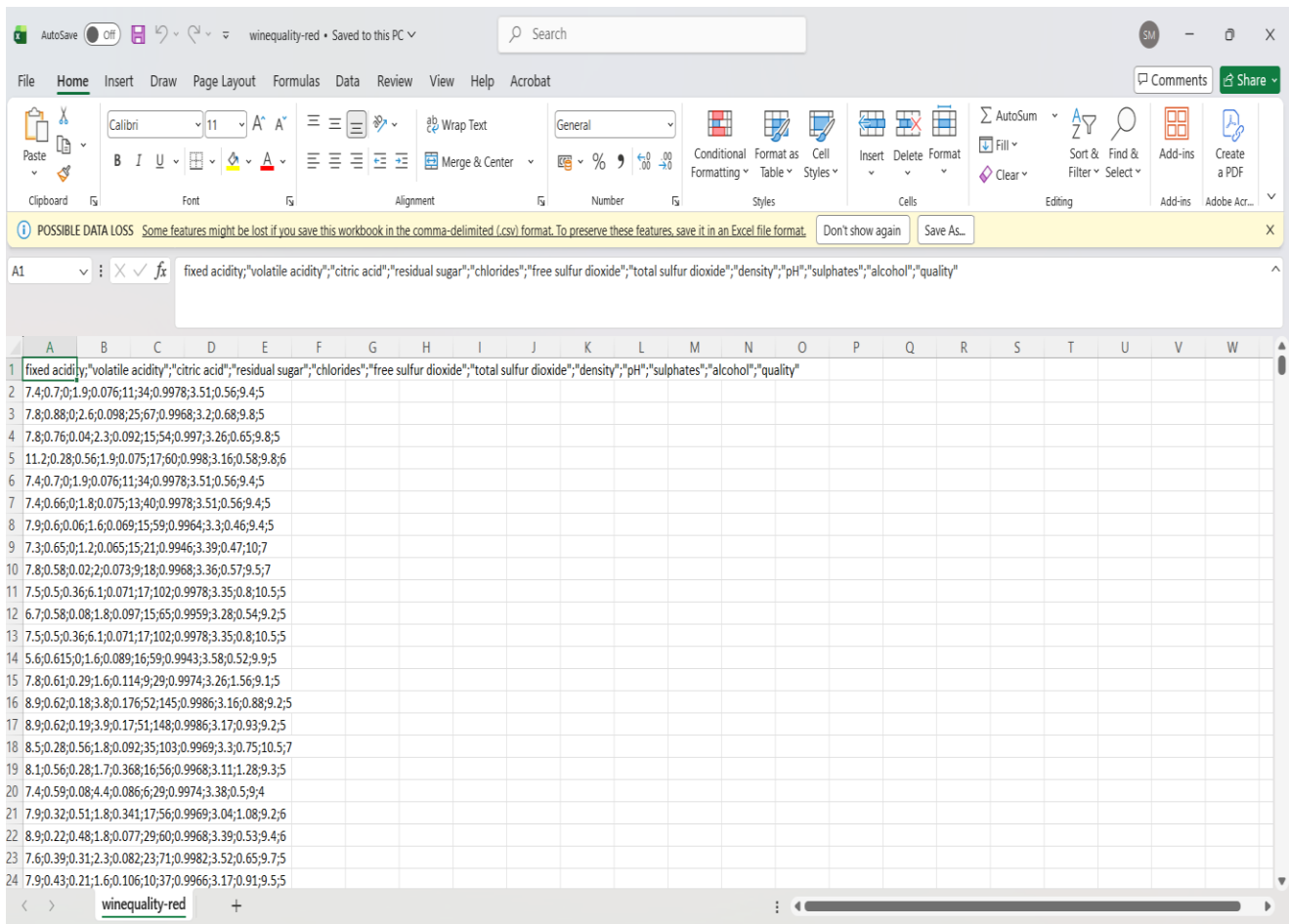
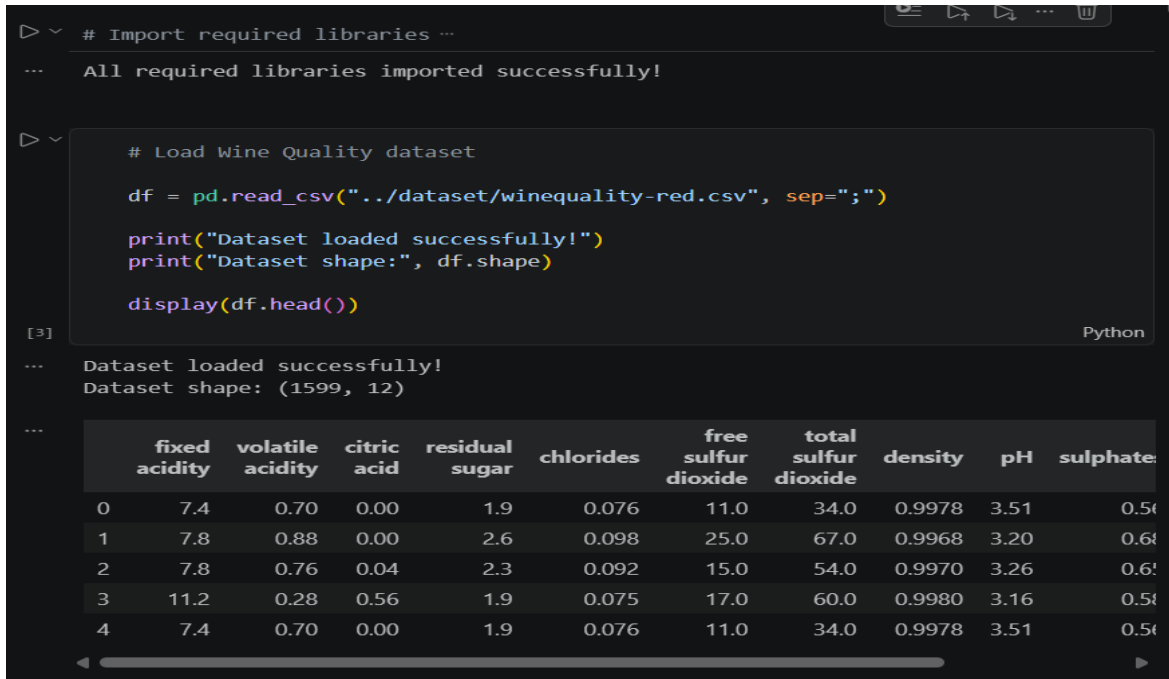


Figure 2.4: Input Dataset – Red Wine Quality Dataset Used for Model Development

CHAPTER 3: WINE QUALITY PREDICTION IMPLEMENTATION

3.1 Library Import & Dataset Loading

The required libraries were imported successfully, including Pandas, NumPy, Matplotlib, Seaborn, train-test splitting utilities, StandardScaler, Random Forest, SGD Classifier, SVC, and classification evaluation metrics. The wine-quality CSV was then loaded into a Pandas DataFrame using `pd.read_csv()` with a semicolon separator.



```
# Import required libraries ...
... All required libraries imported successfully!

# Load Wine Quality dataset

df = pd.read_csv("../dataset/winequality-red.csv", sep=";")

print("Dataset loaded successfully!")
print("Dataset shape:", df.shape)

display(df.head())
```

[3] Python

Dataset loaded successfully!
Dataset shape: (1599, 12)

	fixed acidity	volatile acidity	citric acid	residual sugar	chlorides	free sulfur dioxide	total sulfur dioxide	density	pH	sulphate
0	7.4	0.70	0.00	1.9	0.076	11.0	34.0	0.9978	3.51	0.51
1	7.8	0.88	0.00	2.6	0.098	25.0	67.0	0.9968	3.20	0.61
2	7.8	0.76	0.04	2.3	0.092	15.0	54.0	0.9970	3.26	0.61
3	11.2	0.28	0.56	1.9	0.075	17.0	60.0	0.9980	3.16	0.51
4	7.4	0.70	0.00	1.9	0.076	11.0	34.0	0.9978	3.51	0.51

Figure 3.1: Initial Dataset Preview

Interpretation:

The preview confirms that the dataset contains physicochemical measurements together with the original quality score. The columns are numerical, making them directly suitable for numerical exploratory analysis and classification preprocessing.

3.2 Data Inspection & Quality Assessment

The DataFrame information output was used to inspect the number of records, columns, non-null counts, and data types. The dataset contains 1,599 records and 12 columns. Eleven columns are float64 numerical variables and quality is stored as an integer variable.

```
> ~
# Inspect dataset structure

print("Dataset Information:")
df.info()

[4]
... Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1599 entries, 0 to 1598
Data columns (total 12 columns):
#   Column                Non-Null Count  Dtype
---  ---
0   fixed acidity          1599 non-null   float64
1   volatile acidity       1599 non-null   float64
2   citric acid            1599 non-null   float64
3   residual sugar         1599 non-null   float64
4   chlorides              1599 non-null   float64
5   free sulfur dioxide    1599 non-null   float64
6   total sulfur dioxide   1599 non-null   float64
7   density                1599 non-null   float64
8   pH                    1599 non-null   float64
9   sulphates              1599 non-null   float64
10  alcohol                1599 non-null   float64
11  quality                1599 non-null   int64
dtypes: float64(11), int64(1)
memory usage: 150.0 KB
```

Figure 3.2: Dataset Information and Data Types

Interpretation:

All 1,599 records are non-null across the 12 columns. The fully numerical structure simplifies preprocessing because the feature matrix does not require categorical encoding.

3.3 Descriptive Statistics

Descriptive statistics were generated for all numerical variables using `df.describe()`. The analysis provides count, mean, standard deviation, minimum, quartiles, and maximum values for each chemical property and the original quality score.

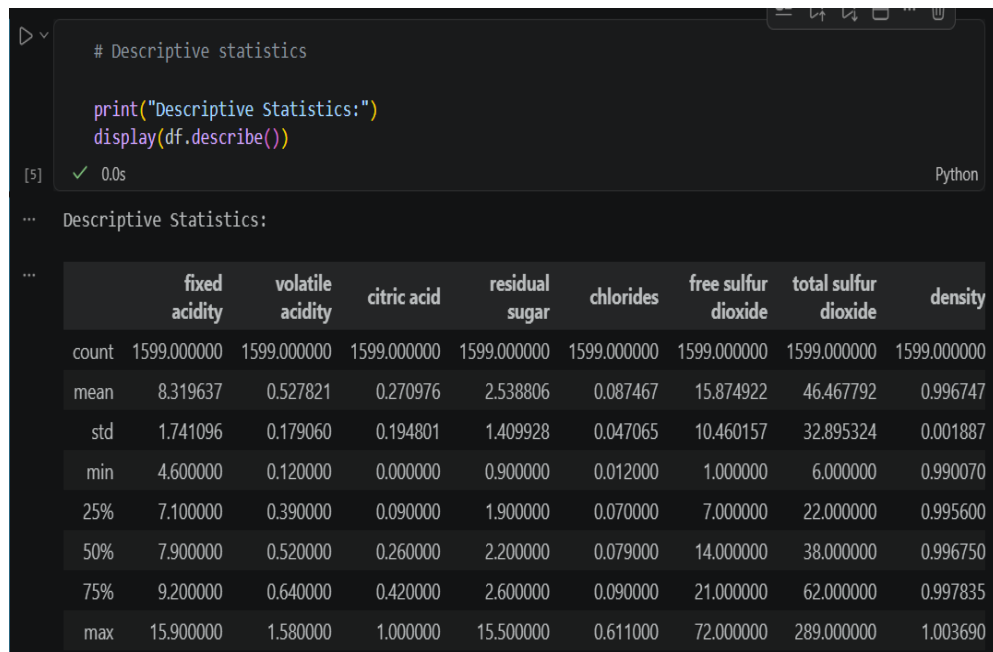


Figure 3.3: Descriptive Statistics of Wine Quality Features

Interpretation:

The variables have different scales and ranges. Alcohol has a mean of approximately 10.42, while quality has a mean of approximately 5.64. The variation in measurement scales supports standardization for scale-sensitive classifiers such as SGD and SVC.

3.4 Missing-Value Analysis

Missing-value analysis was performed using `df.isnull().sum()`. Every feature and the quality column reported a missing count of zero, giving a total of zero missing values across the complete dataset.

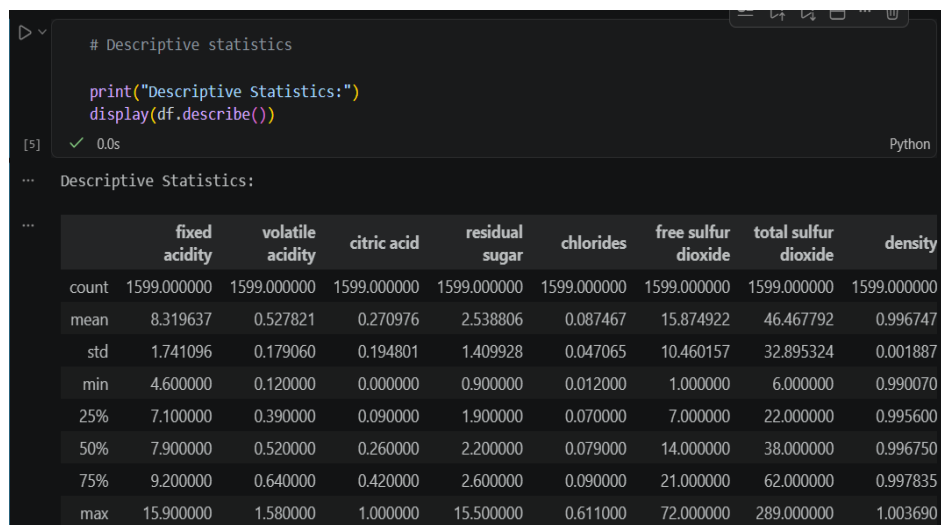


Figure 3.4: Missing-Value Analysis

Interpretation:

The dataset is complete with respect to missing observations. Therefore, no imputation or row deletion was required before modelling.

3.5 Quality Score Distribution

The original quality scores range from 3 to 8. The frequency distribution is concentrated around scores 5 and 6, with 681 wines scored 5 and 638 wines scored 6. Only a small number of wines received scores 3, 4, or 8.

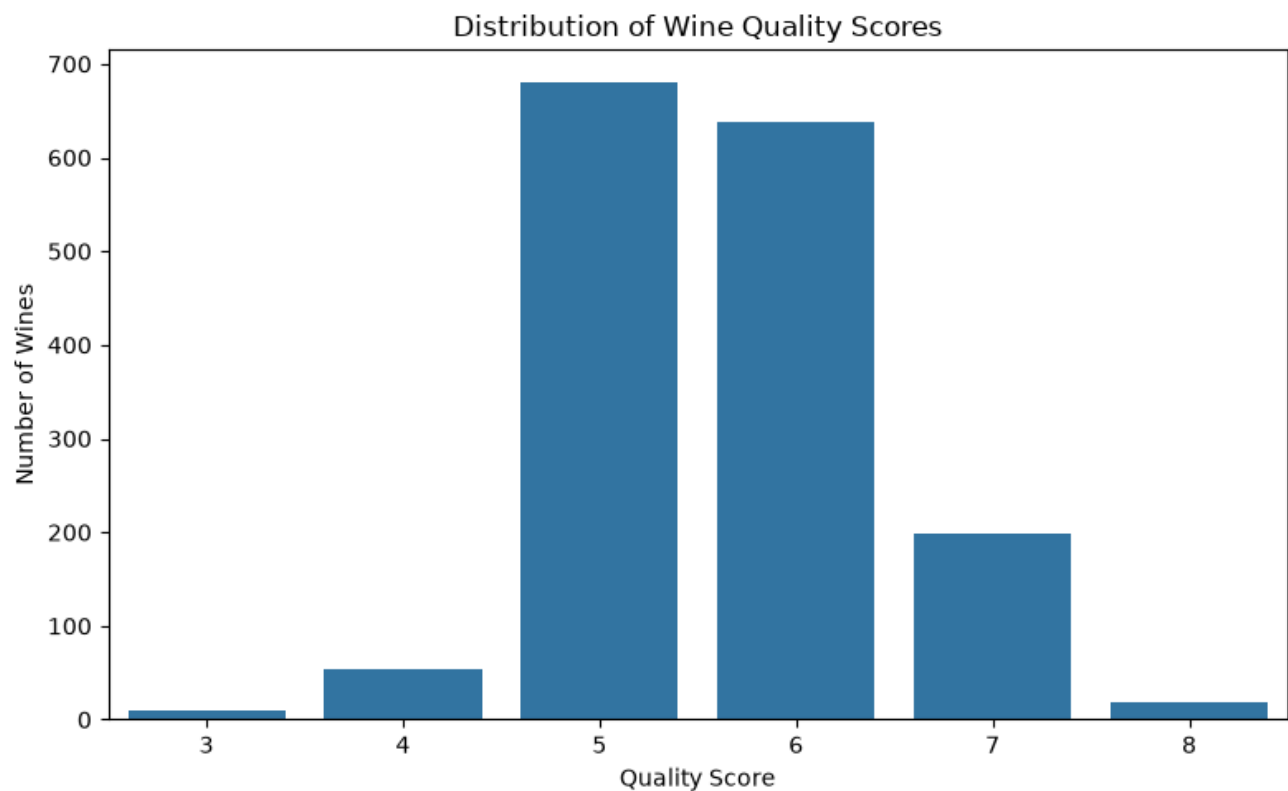


Figure 3.5: Distribution of Wine Quality Scores

Interpretation:

The concentration around scores 5 and 6 indicates that the original target is unevenly distributed across the six observed quality levels. This motivated the practical binary classification used in the project.

3.6 Feature Distribution Analysis

Distribution plots were generated for the eleven chemical features using histograms with kernel-density estimates. The plots provide a visual assessment of the spread and concentration of the physicochemical variables before model training.

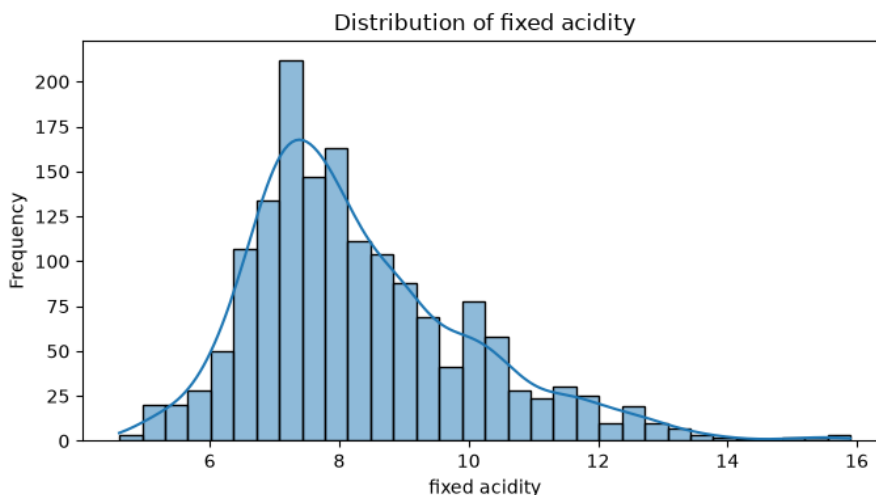


Figure 3.6.1: Distribution of Chemical Feature 1

Interpretation:

The fixed acidity values are mainly concentrated in the lower-to-middle range, with a gradual decline toward higher values. The distribution is positively skewed, indicating relatively fewer observations at high acidity levels.

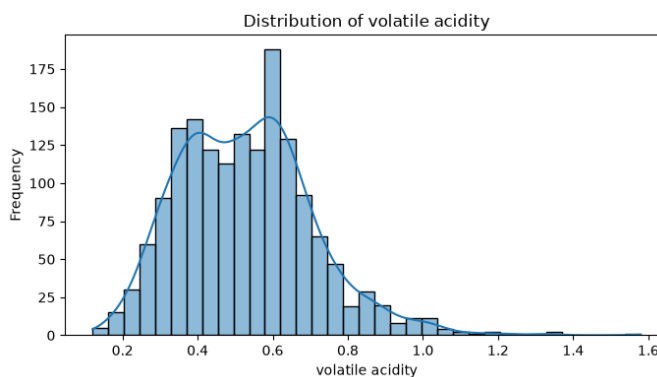


Figure 3.6.2: Distribution of Chemical Feature 2

Interpretation:

Volatile acidity is mostly concentrated around the central range, while higher values occur less frequently. The distribution shows a moderate right skew with a relatively small number of high-value observations.

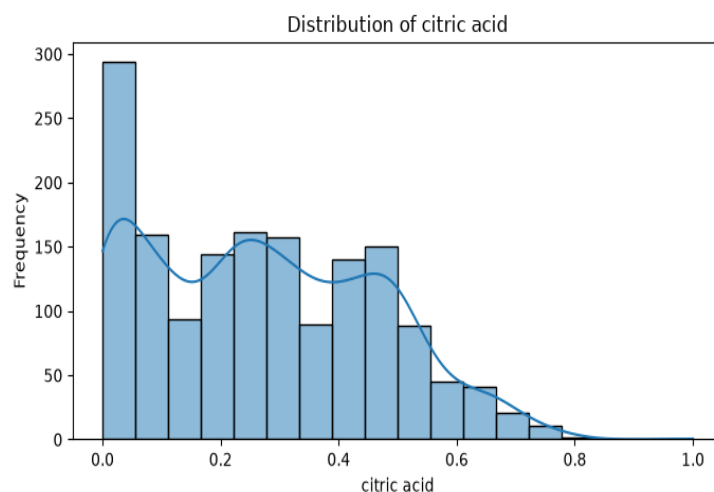


Figure 3.6.3: Distribution of Chemical Feature 3

Interpretation:

Citric acid shows greater variation across its range, with a high concentration at lower values and decreasing frequencies at higher values. This indicates a non-uniform distribution with some right-skewness.

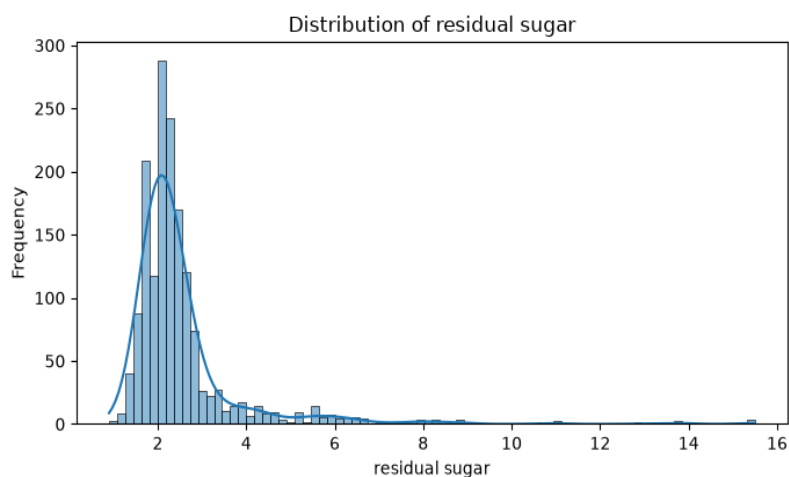


Figure 3.6.4: Distribution of Chemical Feature 4

Interpretation:

Residual sugar is strongly concentrated at lower values and has a long right tail extending to higher values. This indicates substantial positive skewness and the presence of comparatively high-value observations.

3.7 Correlation Analysis

A Pearson correlation matrix was calculated for the complete set of wine-quality variables and visualized as a heatmap. The analysis provides a compact view of both feature-to-feature relationships and relationships between the chemical measurements and the original quality score.

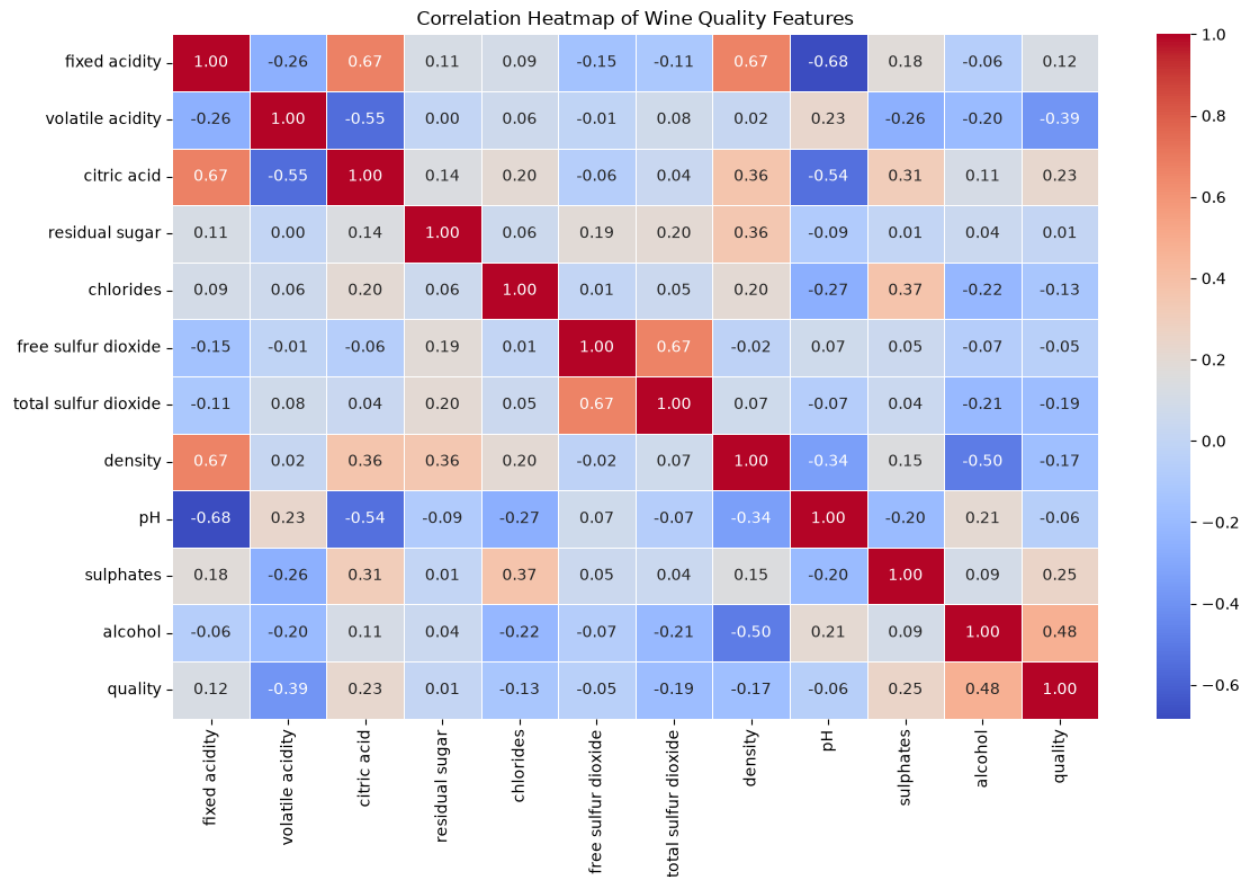


Figure 3.7: Correlation Heatmap of Wine Quality Features

Interpretation:

Alcohol has the strongest positive correlation with quality at approximately 0.48, while volatile acidity has the strongest negative correlation at approximately -0.39. Sulphates also show a positive relationship with quality, while density and total sulfur dioxide show weaker negative relationships. The heatmap also reveals relationships among predictor variables, such as fixed acidity with citric acid and density, and free sulfur dioxide with total sulfur dioxide.

3.8 Binary Quality Classification

The original quality score was converted into a binary target using the rule $\text{quality} \geq 6$. Wines with quality scores below 6 were labelled 0, representing Not Good, while wines with scores of 6 or above were labelled 1, representing Good.

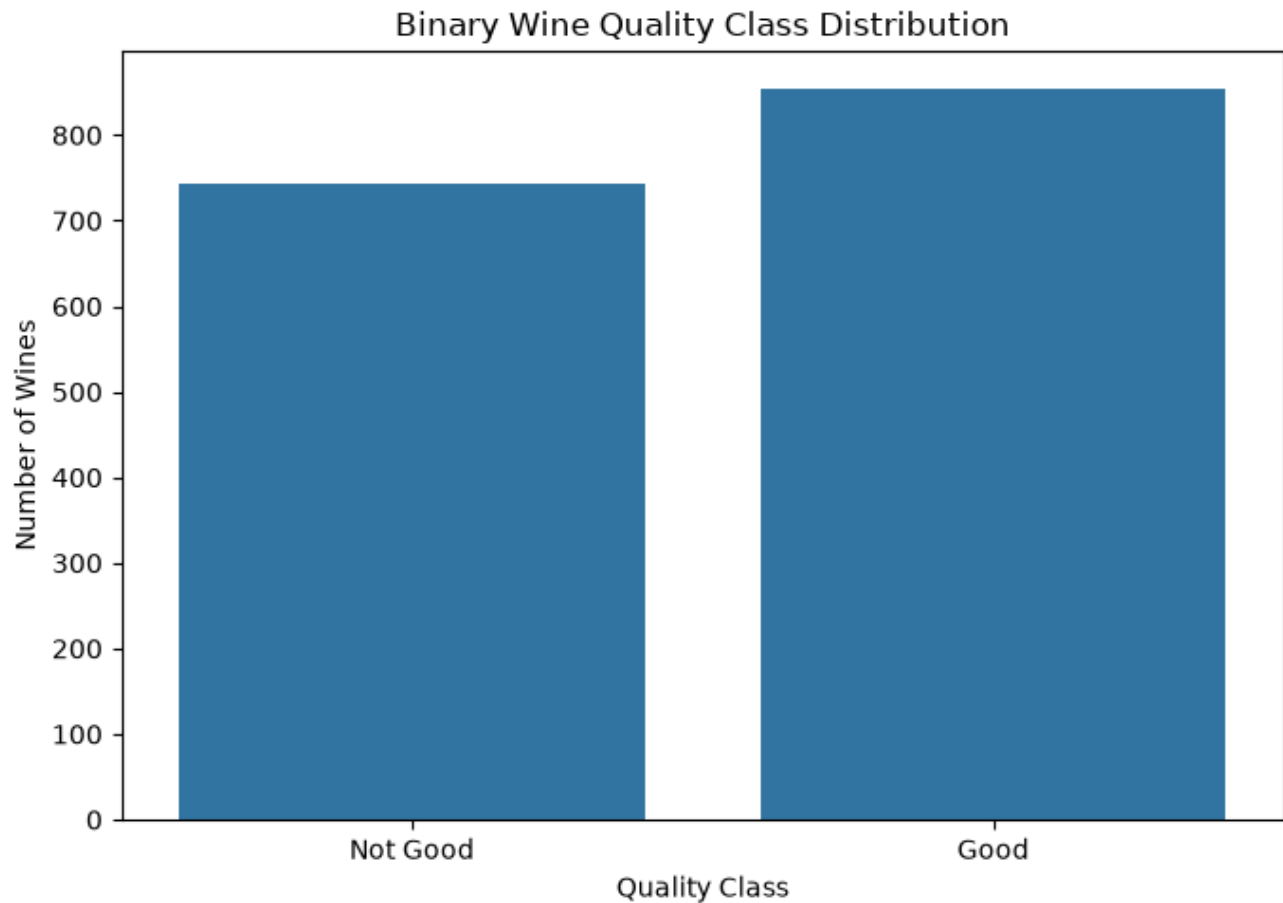


Figure 3.8: Binary Wine Quality Class Distribution

Interpretation:

The transformation produced 744 Not Good wines and 855 Good wines. The resulting classes are relatively balanced, although the Good class is somewhat larger. This class balance is suitable for the planned classification workflow.

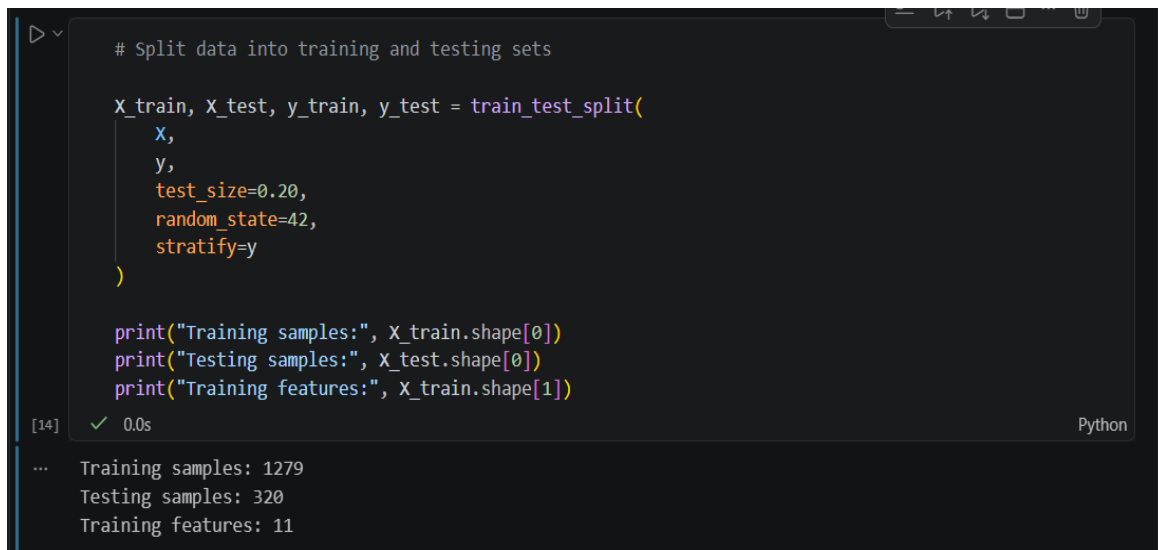
3.9 Feature Preparation

The feature matrix X was created by removing both the original quality score and the newly created quality_label from the predictors. This produced 11 input features. The target vector y contains the binary quality_label.

The separation ensures that the model receives only physicochemical measurements as predictors and does not accidentally use the original quality score as an input feature.

3.10 Train-Test Split

The dataset was divided using `train_test_split()` with `test_size=0.20`, `random_state=42`, and `stratify=y`. The split produced 1,279 training samples and 320 testing samples, with all 11 physicochemical features retained.



```
# Split data into training and testing sets

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42,
    stratify=y
)

print("Training samples:", X_train.shape[0])
print("Testing samples:", X_test.shape[0])
print("Training features:", X_train.shape[1])
```

[14] ✓ 0.0s Python

... Training samples: 1279
Testing samples: 320
Training features: 11

Figure 3.10: Training and Testing Split

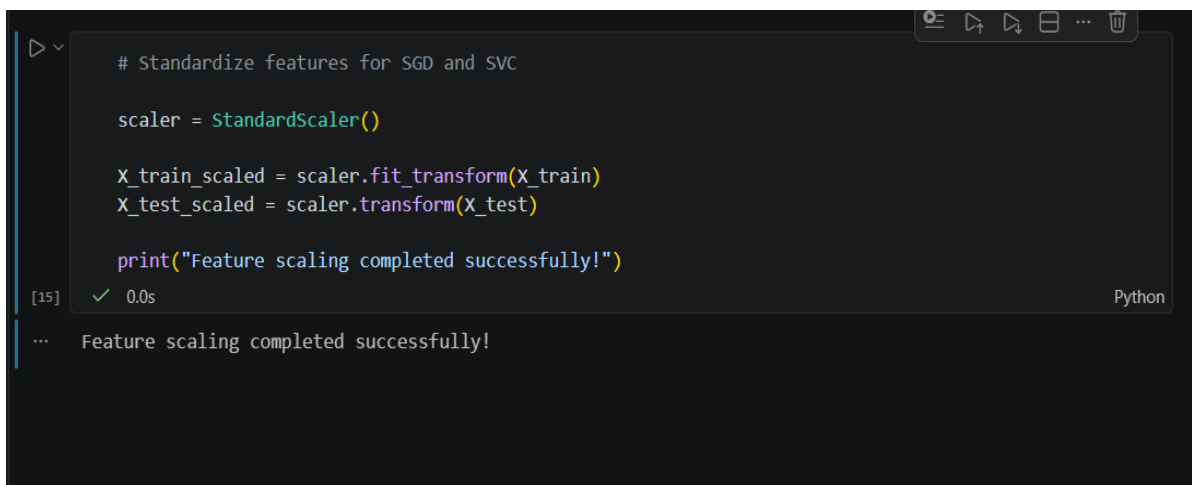
Interpretation:

The 80:20 split provides a separate test set for evaluating generalization. Stratification preserves the Good/Not Good class proportions across the training and testing subsets.

3.11 Feature Scaling

StandardScaler was fitted on the training features and then applied to both the training and testing sets. The scaled features were used for the SGD Classifier and SVC models, which are sensitive to differences in feature magnitude.

Random Forest does not require feature scaling for this implementation, so it was trained directly on the original training feature values.



```
# Standardize features for SGD and SVC

scaler = StandardScaler()

X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

print("Feature scaling completed successfully!")
```

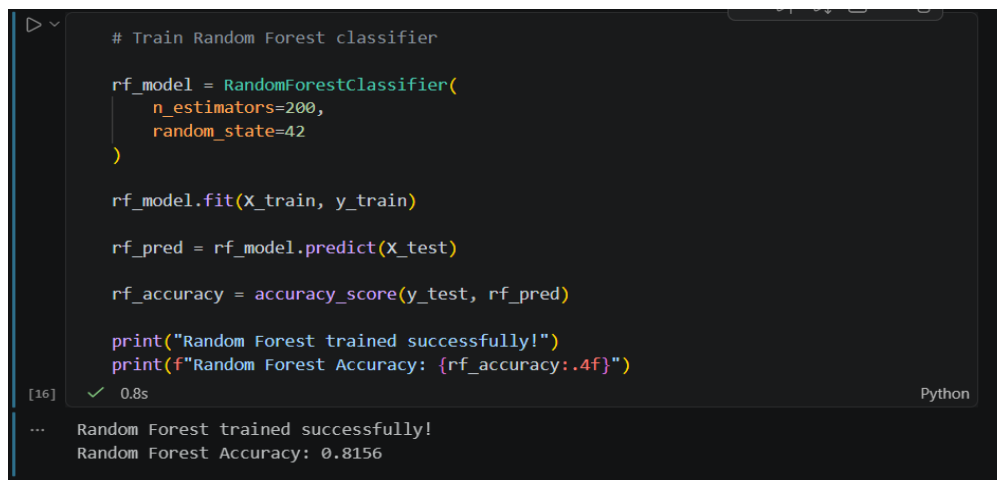
[15] ✓ 0.0s Python

... Feature scaling completed successfully!

Figure 3.11: Feature Scaling Using StandardScaler

3.12 Random Forest Model

A RandomForestClassifier with 200 trees and random_state=42 was trained using the 11 physicochemical features. Predictions were generated for the 320 test observations, and the resulting accuracy was 0.8156.



```
# Train Random Forest classifier

rf_model = RandomForestClassifier(
    n_estimators=200,
    random_state=42
)

rf_model.fit(X_train, y_train)

rf_pred = rf_model.predict(X_test)

rf_accuracy = accuracy_score(y_test, rf_pred)

print("Random Forest trained successfully!")
print(f"Random Forest Accuracy: {rf_accuracy:.4f}")
```

[16] ✓ 0.8s Python

... Random Forest trained successfully!
Random Forest Accuracy: 0.8156

Figure 3.11.1: Classification Model Accuracy Results

Interpretation:

Random Forest produced the highest test accuracy among the three models. Its ensemble of decision trees captured nonlinear relationships between the chemical measurements and the binary quality label effectively.

3.13 Random Forest Evaluation

The Random Forest classifier achieved an accuracy of 81.56%. The classification report shows precision of 0.79 and recall of 0.82 for Not Good wines, and precision of 0.84 and recall of 0.81 for Good wines.



Figure 3.13: Random Forest Classification Report

Interpretation:

The precision and recall values are reasonably balanced between the two classes. The model correctly identifies both Good and Not Good wines without showing a strong bias toward either class.

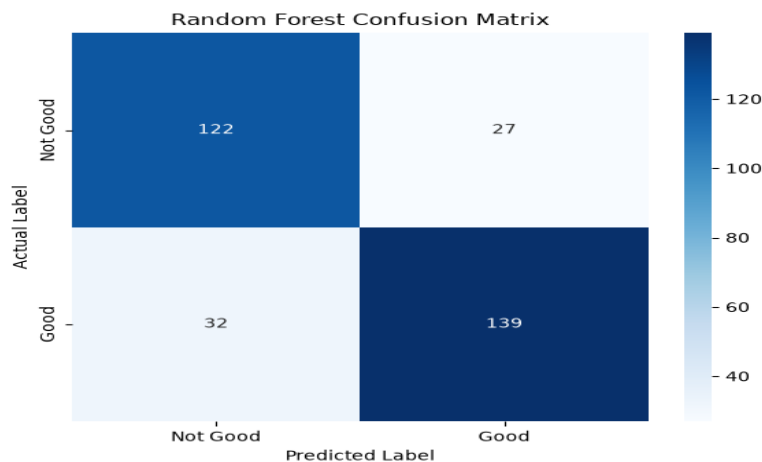


Figure 3.13.1: Random Forest Confusion Matrix

Interpretation:

The confusion matrix provides the class-by-class prediction counts and confirms the overall accuracy reported by the classifier.

3.14 SGD Classifier

A Stochastic Gradient Descent classifier was trained using the standardized training data. The model used `random_state=42`, `max_iter=1000`, and `tol=1e-3`. Its test accuracy was 0.6562, which is substantially lower than the Random Forest result.

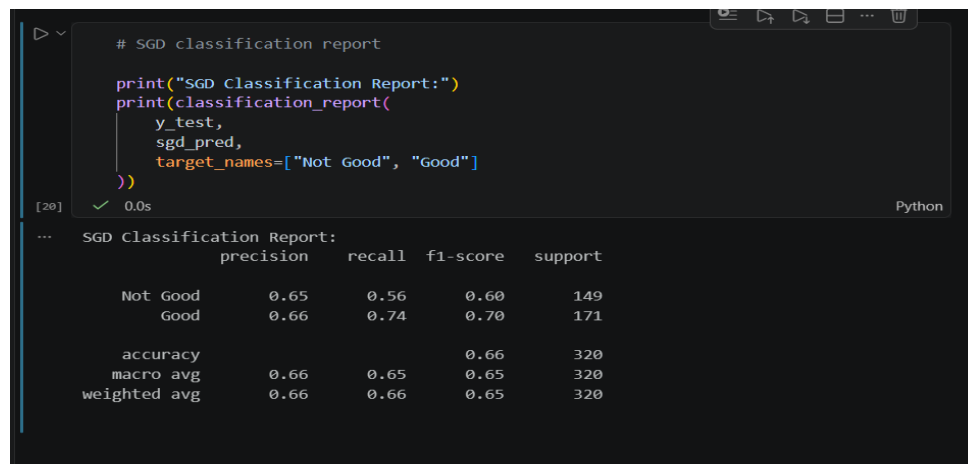


Figure 3.14: SGD Classifier Classification Report

Interpretation:

The SGD classifier achieved 0.66 accuracy. Its recall for the Good class was 0.74, but its recall for Not Good was only 0.56, indicating that the model was less consistent in identifying the Not Good class.

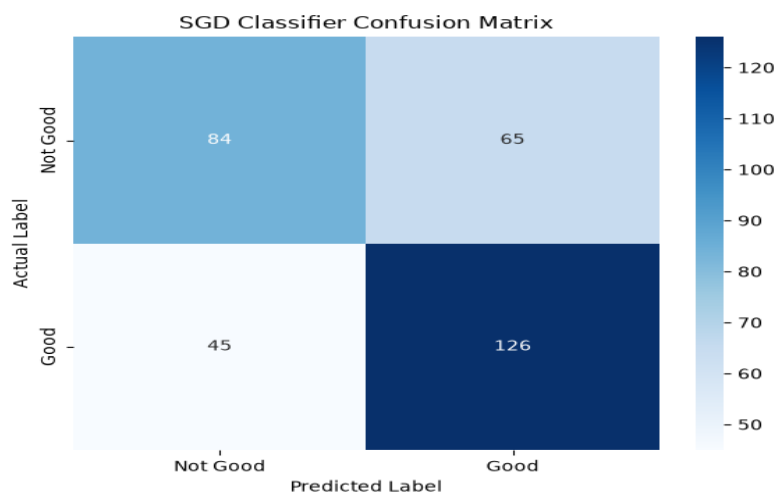


Figure 3.14.1: SGD Classifier Confusion Matrix

Interpretation:

The confusion matrix illustrates the larger number of classification errors made by SGD compared with Random Forest.

3.15 Support Vector Classifier (SVC)

A Support Vector Classifier with an RBF kernel was trained on the standardized features. The model achieved a test accuracy of 0.7625.

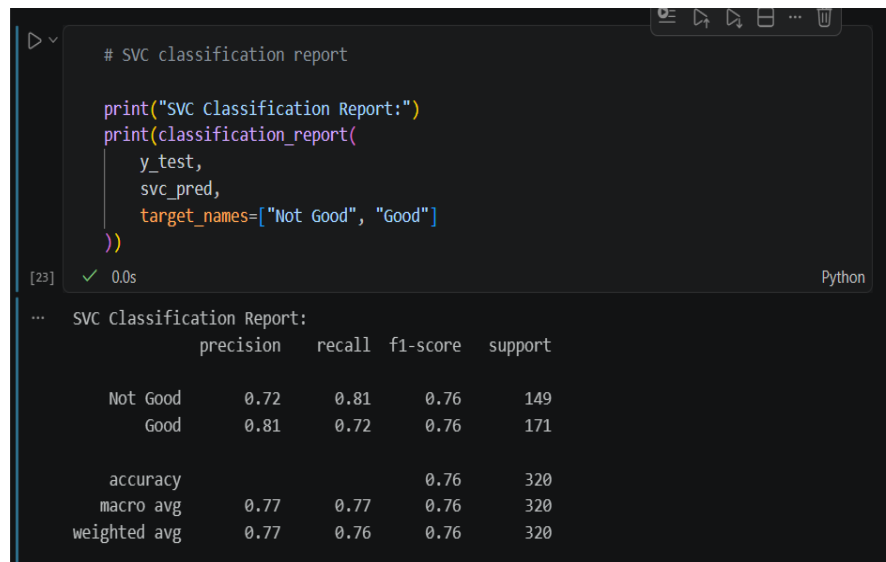


Figure 3.15: SVC Classification Report

Interpretation:

SVC achieved 76.25% accuracy, placing it between Random Forest and SGD. Precision and recall were balanced at approximately 0.72–0.81 for the two classes.

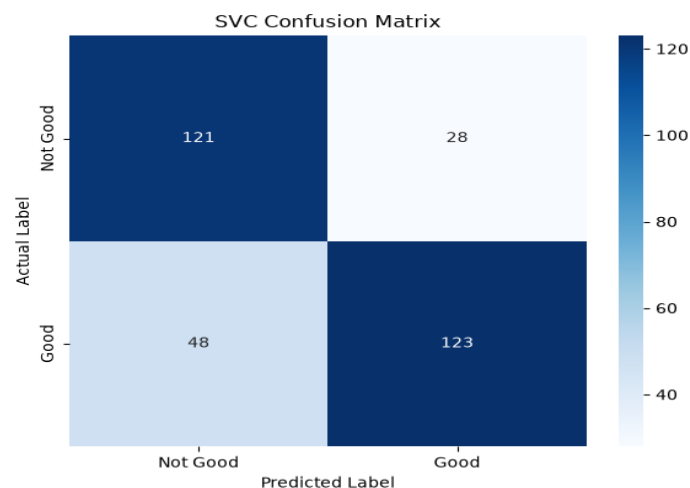


Figure 3.15.1: SVC Confusion Matrix

Interpretation:

The confusion matrix indicates that SVC made fewer errors than SGD but more errors than Random Forest on the same test set.

3.16 Model Comparison

The three classifiers were compared using their test-set accuracy. Random Forest achieved 81.56%, SVC achieved 76.25%, and SGD Classifier achieved 65.63%. The comparison was performed using the same target definition and the same stratified train-test split.

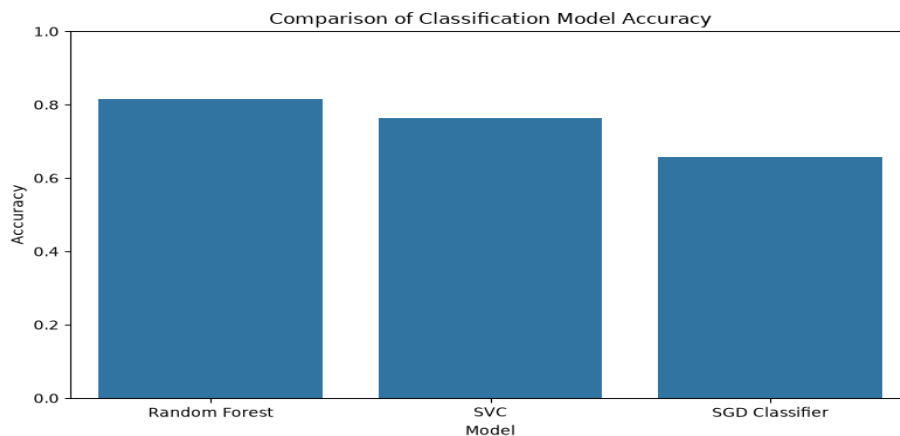


Figure 3.16: Comparison of Classification Model Accuracy

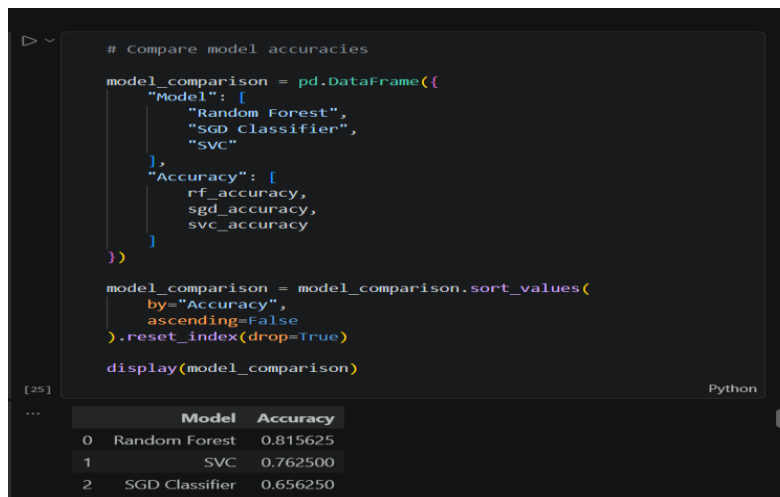


Figure 3.16.1: Numerical Model Accuracy Comparison

Interpretation:

Random Forest is the best-performing model among the three evaluated classifiers. Its accuracy is approximately 5.31 percentage points higher than SVC and 15.94 percentage points higher than SGD Classifier on the test set.

3.17 Random Forest Feature Importance

Feature importance was extracted from the trained Random Forest model to identify which physicochemical variables contributed most strongly to its classification decisions. Alcohol had the highest importance at 0.178138, followed by sulphates at 0.138765 and volatile acidity at 0.113982.

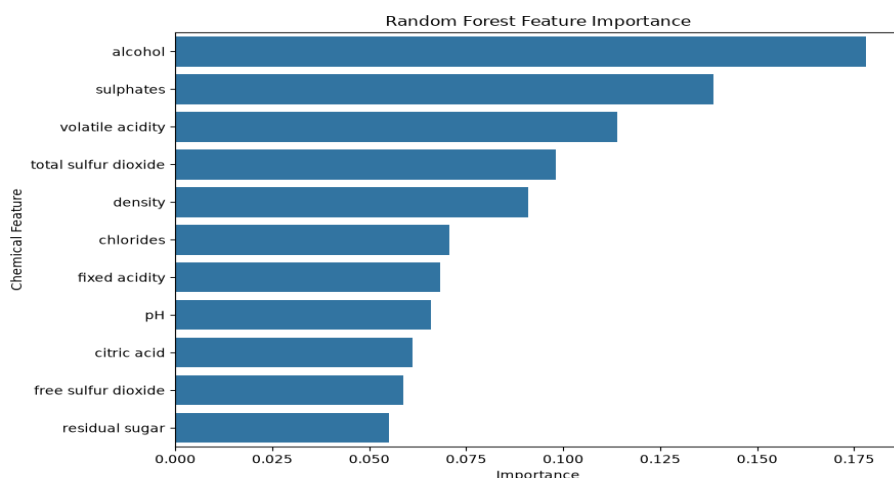


Figure 3.17: Random Forest Feature Importance

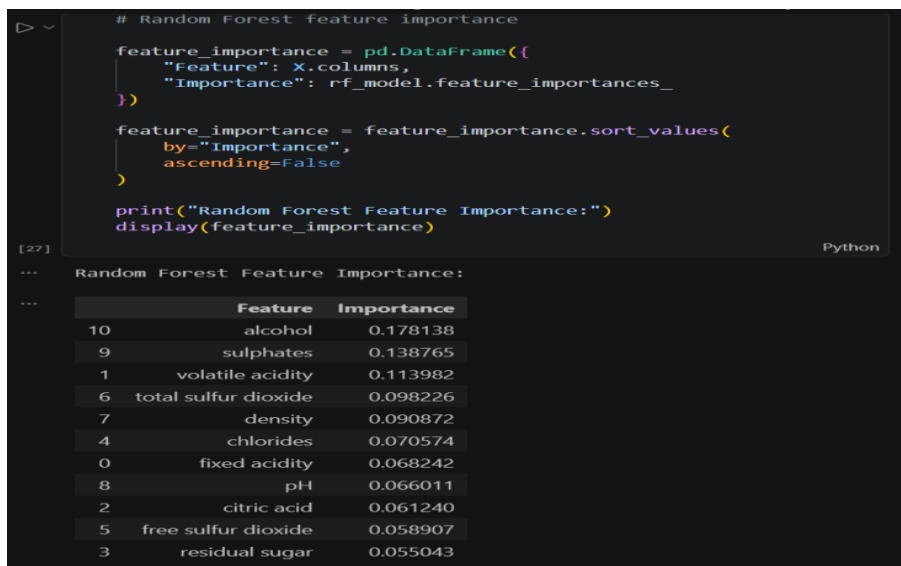


Figure 3.17.1: Numerical Random Forest Feature Importance

Interpretation:

Alcohol is the most influential feature in the fitted Random Forest, followed by sulphates and volatile acidity. The result is consistent with the earlier correlation analysis, where alcohol showed the strongest positive relationship with the original quality score and volatile acidity showed the strongest negative relationship.

CHAPTER 4: RESULTS AND DISCUSSION

4.1 Key Findings

METRIC	RESULT
Initial dataset	1,599 rows $\times$ 12 columns
Input features	11 physicochemical features
Missing values	0
Binary target	Not Good < 6 ; Good ≥ 6
Training / testing split	1,279 / 320 samples
Random Forest accuracy	81.56%
SVC accuracy	76.25%
SGD accuracy	65.63%
Best model	Random Forest
Strongest positive correlation	Alcohol ≈ 0.48
Strongest negative correlation	Volatile acidity ≈ -0.39
Most important RF feature	Alcohol = 0.178138

Interpretation:

The results demonstrate that the project completed the required classification workflow. The dataset was inspected and found to contain no missing values, the original quality score was converted into a practical binary target, and three classification algorithms were evaluated consistently. Random Forest produced the strongest test performance.

4.2 Conclusion

The Level 2 – Task 2 Wine Quality Prediction project was successfully completed using Python and Scikit-learn. The workflow systematically loaded the red wine quality dataset, examined its structure and distributions, verified data quality, analyzed correlations, constructed a binary classification target, split the data using stratified sampling, standardized features for scale-sensitive models, trained three classifiers, and evaluated their results.

Random Forest achieved the highest test accuracy of 81.56%, followed by SVC at 76.25% and SGD Classifier at 65.63%. Feature-importance analysis further showed that alcohol, sulphates, and volatile acidity were the most influential variables in the Random Forest model. Therefore, within this implementation and test split, Random Forest is the preferred classifier among the evaluated alternatives.

4.3 Future Enhancements

1. Experiment with alternative quality thresholds and multiclass prediction of the original quality scores.
2. Apply hyperparameter tuning and cross-validation to improve the robustness of model comparisons.
3. Evaluate additional classifiers such as Gradient Boosting, XGBoost, or calibrated ensemble models.
4. Investigate class-weighting or resampling strategies if class imbalance becomes more pronounced for alternative target definitions.
5. Perform additional feature engineering and investigate nonlinear interactions among physicochemical variables.
6. Use probability calibration and ROC-AUC, precision-recall curves, and other threshold-based metrics for deeper evaluation.
7. Develop an interactive dashboard or deployment interface for practical wine-quality classification.

4.4 References

1. Wine Quality dataset and the submitted Level 2 – Task 2 implementation notebook.
2. Scikit-learn functionality used for train-test splitting, StandardScaler, Random Forest, SGD Classifier, SVC, accuracy, classification reports, and confusion matrices.
3. Python, Pandas, NumPy, Matplotlib, Seaborn, and Jupyter Notebook used for data preparation, exploratory analysis, visualization, modelling, and documentation.

4.5 Final Findings and Conclusion

The final implementation confirms that physicochemical properties can be used to classify red wines into Good and Not Good quality categories. The dataset contained 1,599 observations with no missing values, and the binary transformation produced 744 Not Good and 855 Good observations.

The correlation analysis identified alcohol as the strongest positive correlate of the original quality score and volatile acidity as the strongest negative correlate. The classification experiments showed clear differences between the three models. Random Forest achieved 81.56% accuracy, SVC achieved 76.25%, and SGD Classifier achieved 65.63%.

The Random Forest feature-importance analysis identified alcohol, sulphates, and volatile acidity as the most influential input variables. Taken together, the exploratory analysis, model evaluation, and feature-importance results provide a complete and interpretable machine-learning workflow for the assigned Wine Quality Prediction task.

Overall, the project successfully met its objectives and demonstrates practical application of data analytics, visualization, supervised classification, model evaluation, and feature interpretation.